Understanding the Loss of Plasticity in Deep Neural Network

Hojoon Lee

2023.11.10

Definition

Machine Learning

- Building a model (M) that learn from the data to generalize to unseen data.

Information Capacity

- A model's capacity to store the information = $\mathrm{H}(\mathrm{M})$.
- Physical capacity = number of parameters = P.
- Information capacity $\leq$ Physical capacity ($\mathrm{H}(\mathrm{M}) \leq \mathrm{P}$).

Generalization

- Having a low generalization error = high performance on unseen data.

Memorization

- Model learns to make predictions based on the noise or random labels.

Definition

Continual Learning

- Building a model that learn from a continual stream of data to generalize to unseen data.
- Here, information capacity = model capacity + buffer capacity.

Stability

- Model's ability to not forget the learned information (data is compressed into the information).
- Catastrophic forgetting = Model forgets useful information from memory.

Plasticity

- Model's ability to adapt to new data.

Stability-Plasticity Dilemma

- Under finite information capacity, increasing stability decreases plasticity and vice-versa.
- Infinite information capacity can resolve this dilemma as we do not need to forget information.

Definition

Loss of Plasticity

- Recently, several researchers report the loss of plasticity in continual learning setup.
- In refers to the model's tendency to lose its ability to adapt to new dataset.
- What's the meaning of losing its ability to adapt?
  1. Fails to minimize the training loss (training accuracy).
  2. Fails to minimize the test loss (test accuracy).
  - Although these two definitions are not identical, they are used interchangeably.
  - I personally think second term is more correct as the purpose of learning is to generalize.

- Surprisingly, even with a large model and infinite size buffer, the loss of plasticity persists.

Contents (but not limited to)

On Warm-Starting Neural Network Training., NeurIPS 2020.

- First observation of the loss of plasticity in terms of generalization.
- The test performance decreases when the model is pre-trained on the half of the training dataset.

Loss of Plasticity in Deep Continual Learning., CoLLA 2022 talk.

- In-depth investigation of the loss of plasticity in terms of trainability (not generalization).
- Found that the loss of plasticity is amplified if the number of continual tasks increases.
- Found that existing regularization techniques were not sufficient to mitigate the loss of plasticity.

Curvature explains Loss of Plasticity., submitted to ICLR 2024.

- Attempts to explain the fundamental cause of the loss of plasticity in terms of trainability.
- Found that the curvature of loss surface explains the loss of plasticity (limited to synthetic experiments).

On Warm-Starting Neural Network Training

Motivation

- Continual Learning is essential as the data and model gets larger and bigger.
- Can we utilize the pre-trained model (=warm-start model) as the prior?

Experimental Setup

- Dataset: CIFAR-10, CIFAR-100, SVHN
- Model: Resnet-18
- Training:
  - Split Dataset into two chunks: A, B
  - Train the model, θ, from the chunk A.
  - Continually train the model, θ, from the chunk A & B.

On Warm-Starting Neural Network Training

Results

CIFAR-10	ResNet SGD	ResNet Adam	MLP SGD	MLP Adam	LR SGD	LR Adam
Random Init	56.2 (1.0)	78.0 (0.6)	39.0 (0.2)	39.4 (0.1)	40.5 (0.6)	33.8 (0.6)
Warm Start	51.7 (0.9)	74.4 (0.9)	37.4 (0.2)	36.1 (0.3)	39.6 (0.2)	33.3 (0.2)
SVHN						
Random Init	89.4 (0.1)	93.6 (0.2)	76.5 (0.3)	76.7 (0.4)	28.0 (0.2)	22.4 (1.3)
Warm Start	87.5 (0.7)	93.5 (0.4)	75.4 (0.1)	69.4 (0.6)	28.0 (0.3)	22.2 (0.9)
CIFAR-100						
Random Init	18.2 (0.3)	41.4 (0.2)	10.3 (0.2)	11.6 (0.2)	16.9 (0.18)	10.2 (0.4)
Warm Start	15.5 (0.3)	35.0 (1.2)	9.4 (0.0)	9.9 (0.1)	16.3 (0.28)	9.9 (0.3)

Table 1: Validation percent accuracies for various optimizers and models for warm-started and randomly initialized models on indicated datasets. We consider an 18-layer ResNet, three-layer multilayer perceptron (MLP), and logistic regression (LR).

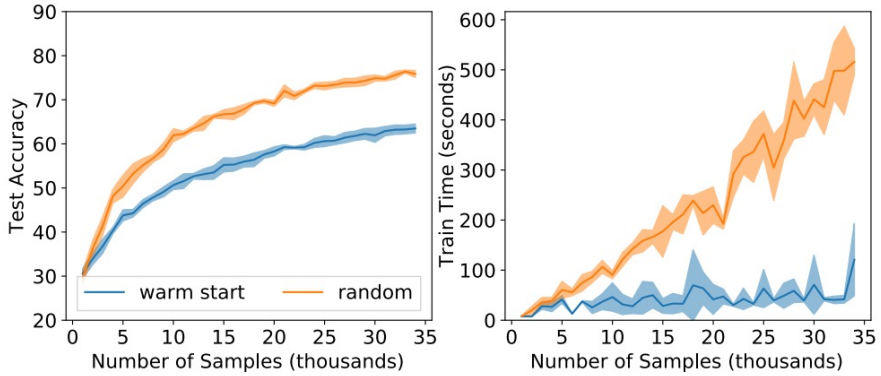


Figure 2: An online learning experiment for CIFAR-10 data using a ResNet. The horizontal axis shows the total number of samples in the training set available to the learner. The generalization gap between warm-started and randomly-initialized models is significant.

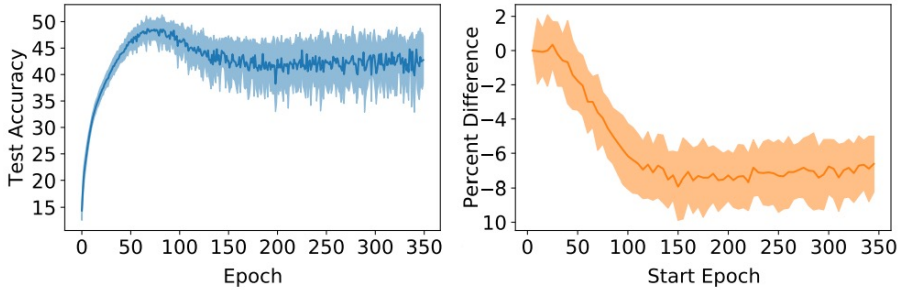


Figure 4: **Left:** Validation accuracy as training progresses on 50% of CIFAR-10. **Right:** Validation accuracy damage, as percentage difference from random initialization, after training on 100% of the data. Each warm-started model was initialized by training on 50% of CIFAR data for the indicated number of epochs.

On Warm-Starting Neural Network Training

Solution: Shrink & Perturb

- When new samples arrive, re-initialize the network parameters by

$$\theta_i^t \leftarrow \lambda \theta_i^{t-1} + p^t, \text{ where } p^t \sim \mathcal{N}(0, \sigma^2) \text{ and } 0 < \lambda < 1.$$

  - (Shrink) Shrinking the pre-trained weights towards zero.
  - (Perturb) Add a gaussian noise to the parameters.

Results of Shrink & Perturb

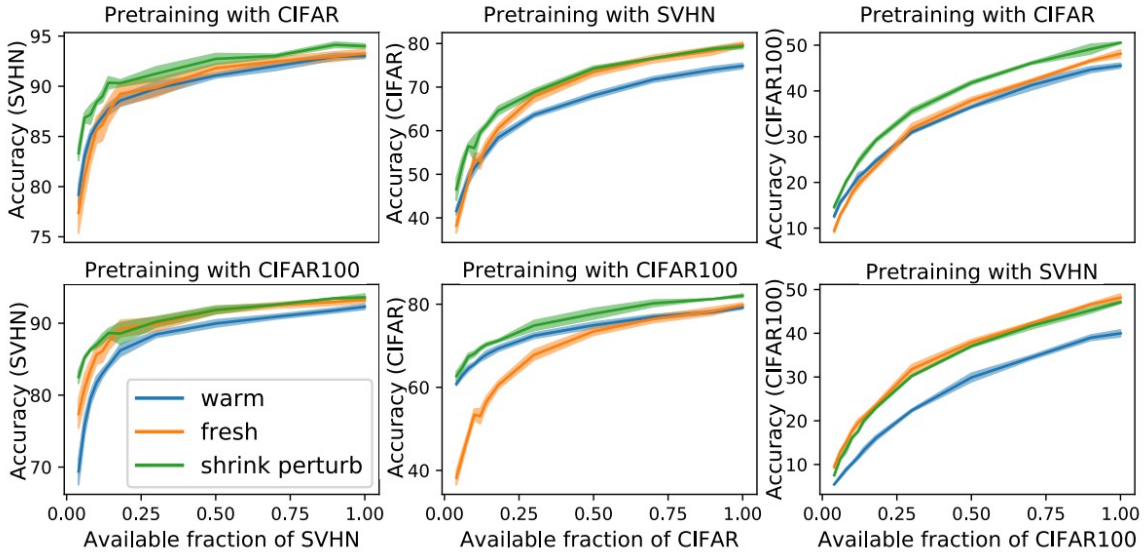


Figure 9: Pre-trained models fitted to a varying fraction of the indicated dataset. We compare these warm-started, pre-trained models to randomly initialized and shrink-perturb initialized counterparts, trained on the same fraction of target data. The relative performance of warm-starting and randomly initializing varies, but shrink-perturb performs at least as well as the best strategy.

On Warm-Starting Neural Network Training

Summary

- Initializing a neural network by first training it on a fraction of the dataset can lead to much lower final performance compared to the case in which the network is trained once on the entire dataset.
- First empirical example of the loss of plasticity (in terms of generalization).
- Shrink & Perturb seems to mitigate this issue in their experimental setup.

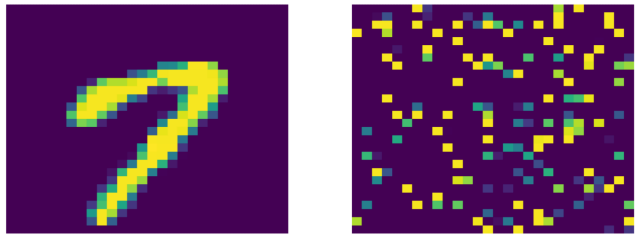
Loss of Plasticity in Deep Continual Learning

Motivation

- More in-depth analysis of the loss of plasticity.
- Designed to understand the loss of plasticity in terms of training.

Experimental Setup

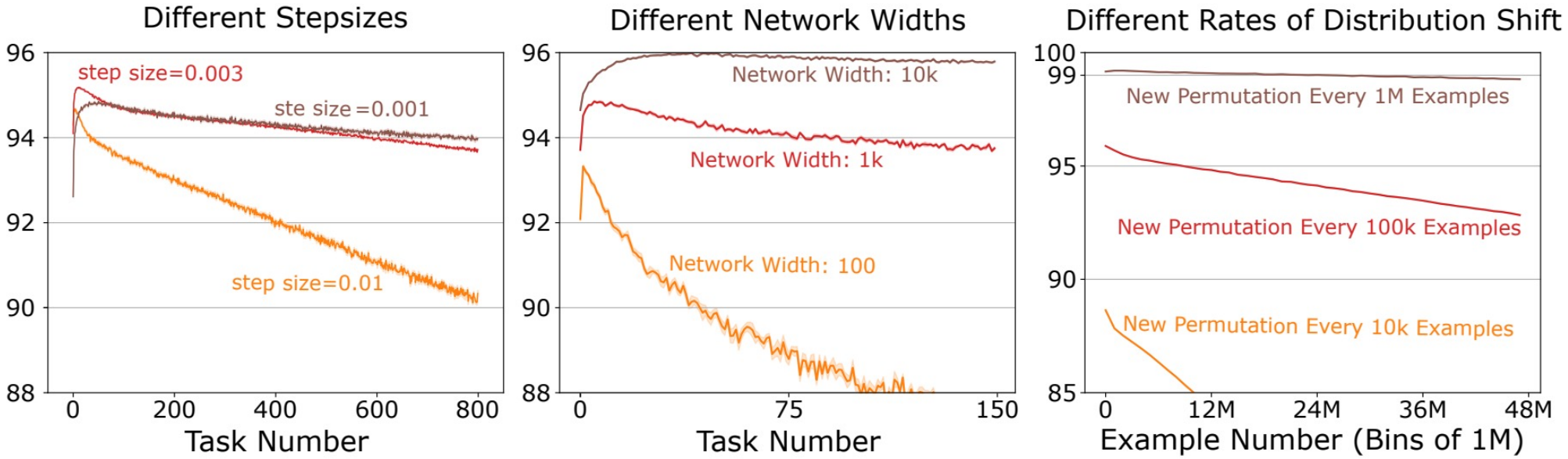
- Dataset: Permuted MNIST
- Model: 3-layer MLP
- Training:
  - For each task, permute the pixels of training dataset.
  - Train the model, θ, from the permuted dataset.
  - Evaluate the train accuracy.



Permuted MNIST

Loss of Plasticity in Deep Continual Learning

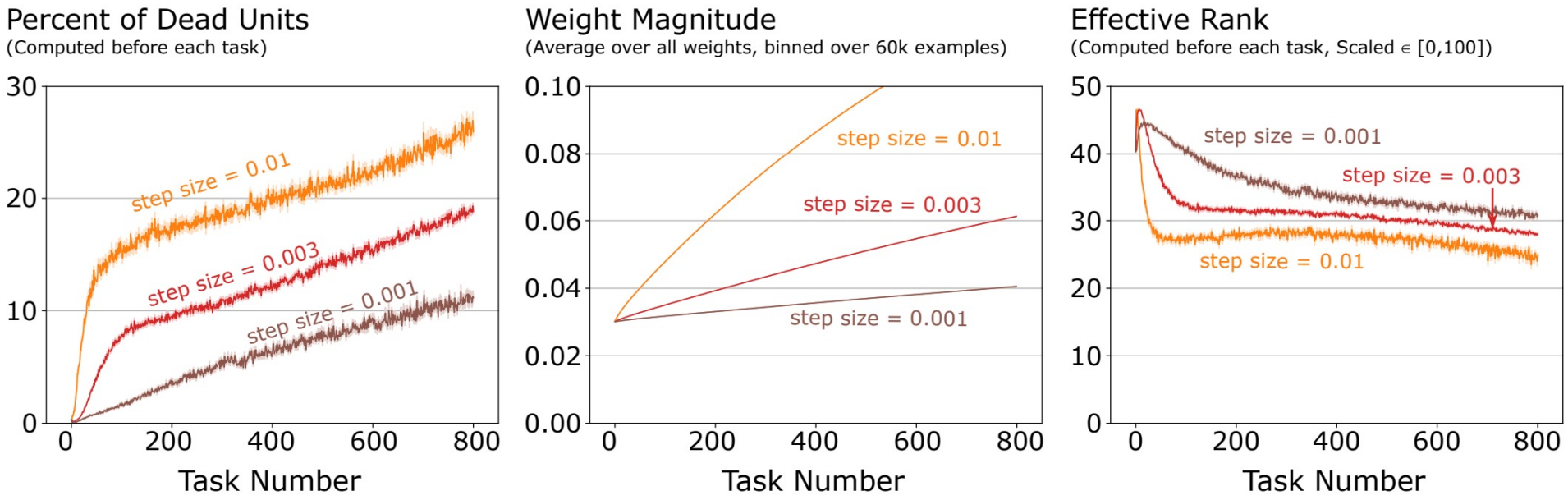
Results



- Loss of plasticity is prevalent when using:
  - Larger step size.
  - Shallow model architecture.
  - Frequent distribution shifts.

Loss of Plasticity in Deep Continual Learning

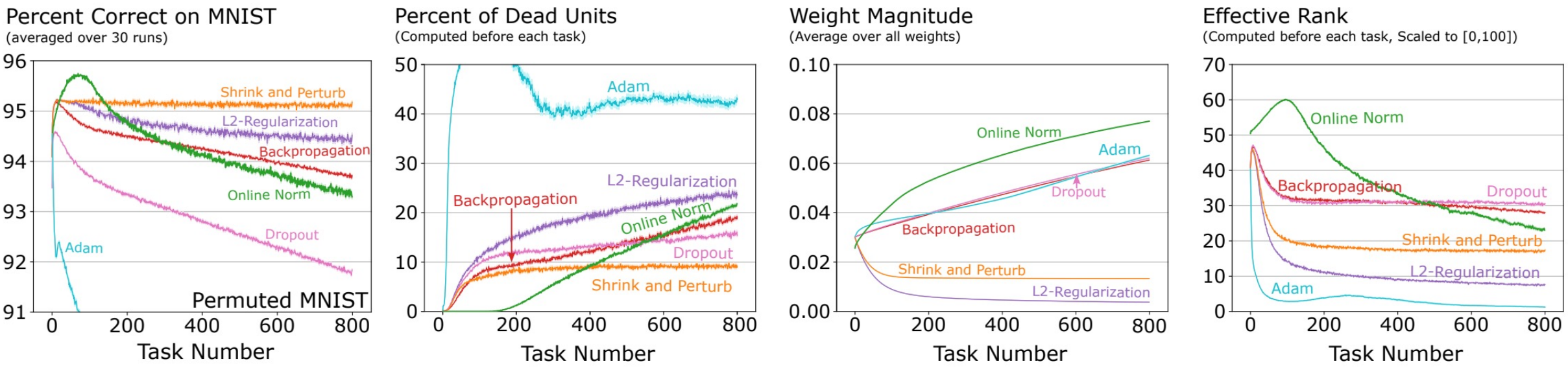
Results



- The reason for the Loss of plasticity may correlated to:
  - The increase of dead units.
  - The increase in weight magnitude.
  - The decrease of the effective feature rank.

Loss of Plasticity in Deep Continual Learning

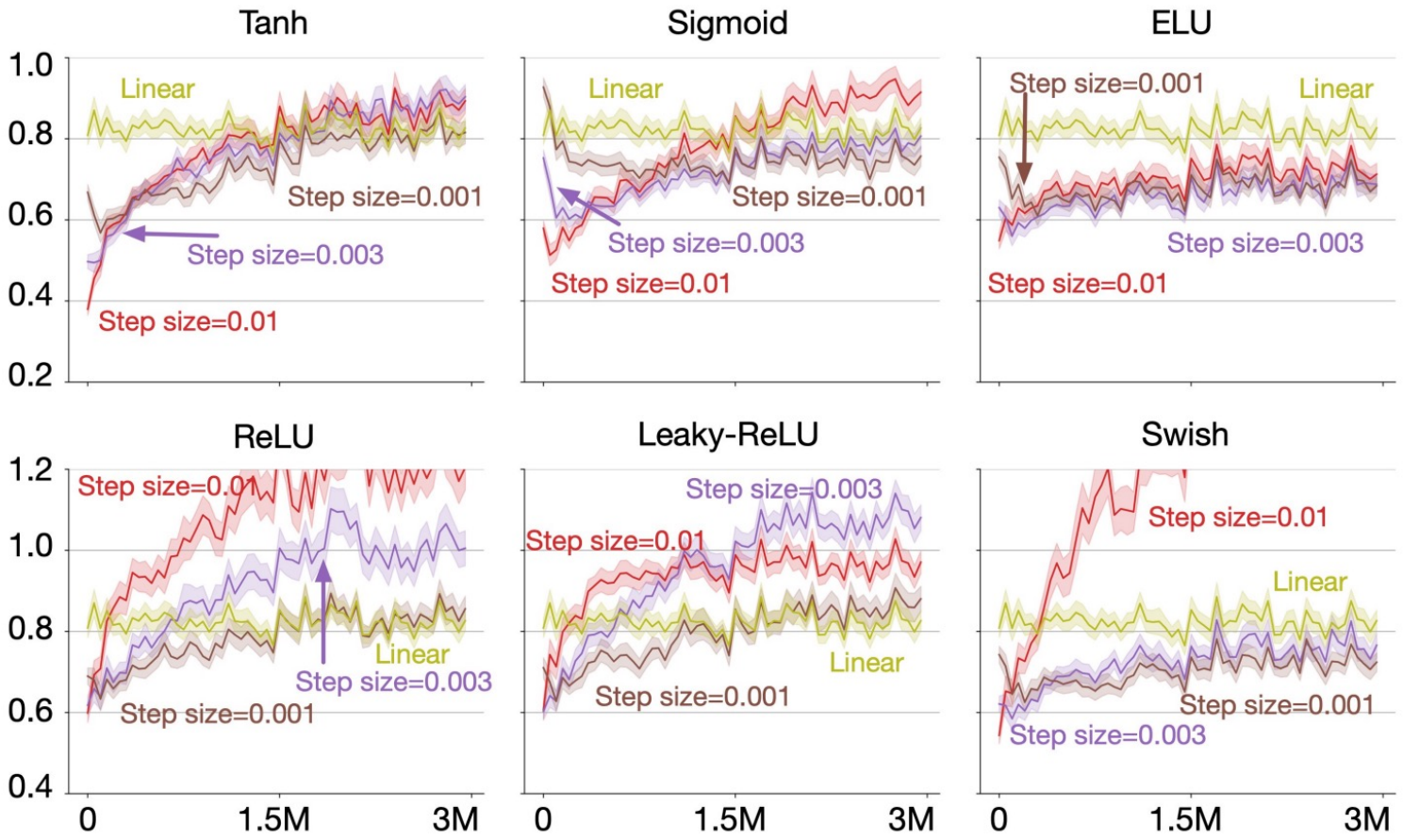
Can we mitigate the loss of plasticity through standard regularization techniques?



- L2-Regularization, Backprop, Dropout do not mitigate the loss of plasticity.
- Shrink & Perturb seems to maintain the model's plasticity.

Loss of Plasticity in Deep Continual Learning

Can we mitigate the loss of plasticity through standard regularization techniques?



- Changing the activation functions (removing dead units) did not resolve the problem.

Loss of Plasticity in Deep Continual Learning

Summary

- First in-depth investigation of the loss of plasticity in terms of trainability (not generalization).
- Found that the loss of plasticity is amplified if the number of continual tasks increases.
- Found that existing regularization techniques were not sufficient to mitigate the loss of plasticity.

Limitations

- Experiments limited to the synthetic datasets with a small model size.
- None of the studied metrics do not explain the loss of plasticity.
- Loss of plasticity, in generalization perspective, is not considered.

Curvature explains Loss of Plasticity

Motivation

- Propose a metric to explain the loss of plasticity phenomenon (in terms of training).
- Found that the curvature of loss surface partially explains the loss of plasticity.

Experimental Setup

- Dataset: Permuted MNIST, Label-Shift MNIST, Permuted CIFAR-10, Label-Shift CIFAR-10
- Model: 3-layer MLP (MNIST), 3-layer Conv + 2-layer MLP (CIFAR-10).
- Training:
  - For each task, permute the pixels / assign random labels for each image in training dataset.
  - Train the model, θ, from the task dataset.
  - Evaluate the train accuracy.

Curvature explains Loss of Plasticity

Does existing metrics explain the loss of plasticity?

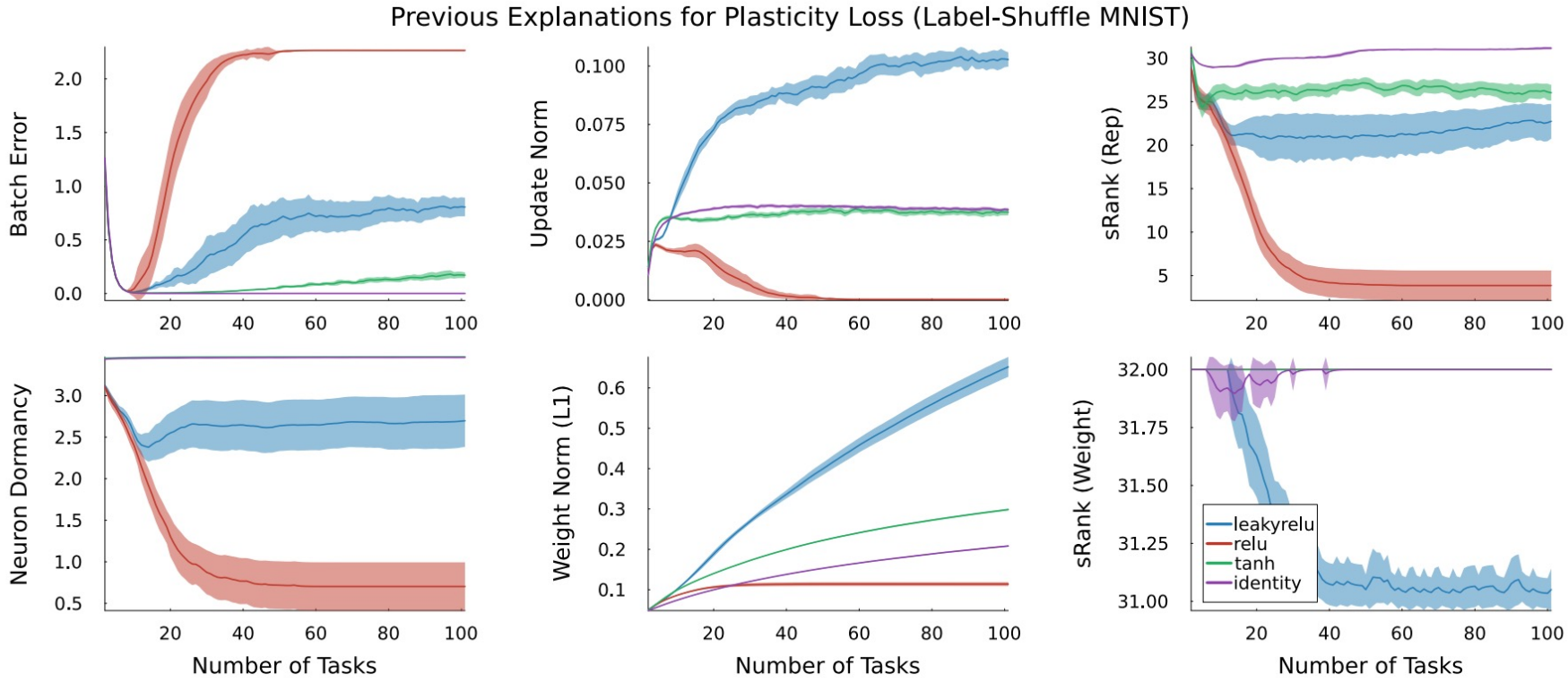


Figure 1: All results have a shaded region corresponding to a 95% confidence interval of the mean. Top left: All non-linear activation functions lose plasticity, but a deep linear network with identity activations does not. Rest: None of the aformentioned explanations explain the difference between plasticity loss of different activation functions. For neuron dormancy and the weight stable rank, some activation functions overlap and are not visible.

Curvature explains Loss of Plasticity

Proposed metric: Rank of the Hessian

- 2nd order approximation of Loss curvature: Hessian.

  $$H_t(\theta) = \nabla_\theta^2 J_t(\theta)\big|_{\theta=\theta_t}$$

- Which characteristics of Hessian are we interested in?
  - Rank: $srank\big(H(\theta)\big)$, $\mathbf{srank}(M) = \min\left\{j : \frac{\sum_{i=1}^{j}\sigma_i(M)}{\sum_{i=1}^{d}\sigma_i(M)} > 0.99\right\}$
  - Why? It represents the number of the gradient directions to minimize the loss curvature.
- Approximating Hessian requires the massive computational overhead.
  - Use layer-wise or partial-block wise approximation to minimize its overhead.
  - Then, compute its rank: $srank\big(H_{Block}(\theta)\big)$,
  - In this paper, the authors measures the rank of the last two-layers.

Curvature explains Loss of Plasticity

Results

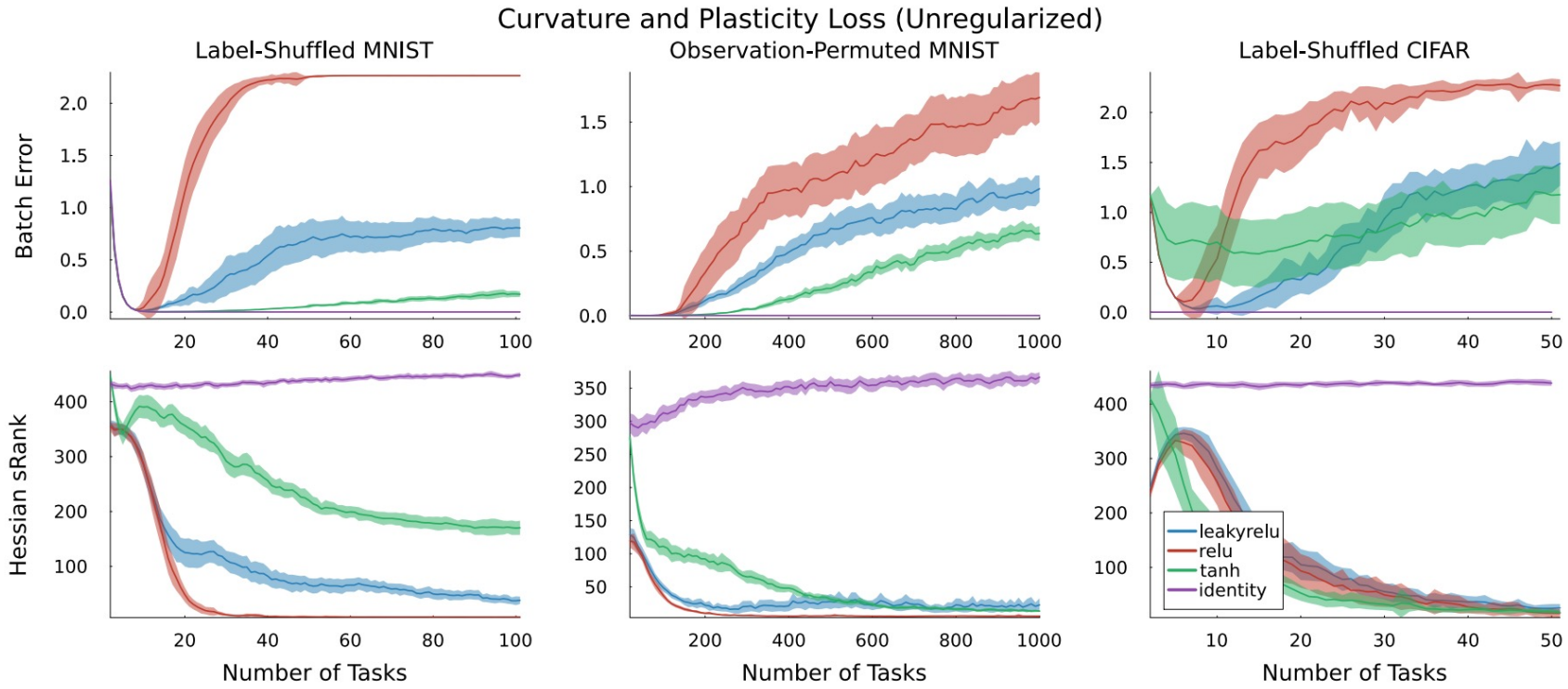


Figure 2: Results with unregularized objectives. Top Row: Batch Error at task end. Bottom Row: sRank of Hessian at task-beginning. Left: Label-shuffled MNIST. Middle: Observation-permuted MNIST. Right: Label-shuffled CIFAR-10.

- Rank of the partial-block wise Hessian succeeds to explain the loss of plasticity.

Take Home Message

On Warm-Starting Neural Network Training., NeurIPS 2020.

- First observation of the loss of plasticity in terms of generalization.
- The test performance decreases when the model is pre-trained on the half of the training dataset.

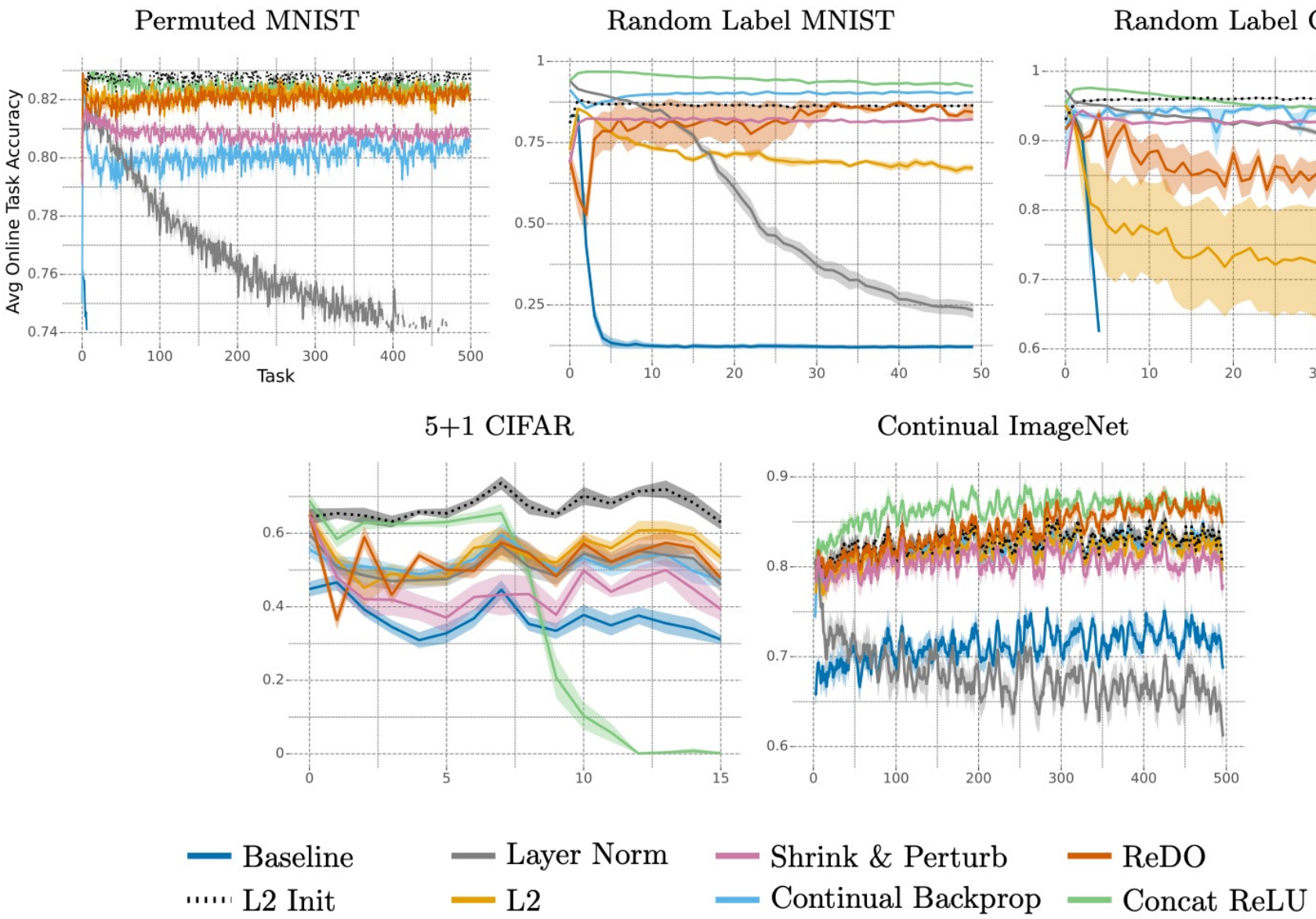
Loss of Plasticity in Deep Continual Learning., CoLLA 2022 talk.

- In-depth investigation of the loss of plasticity in terms of trainability (not generalization).
- Found that the loss of plasticity is amplified if the number of continual tasks increases.
- Found that existing regularization techniques were not sufficient to mitigate the loss of plasticity.

Curvature explains Loss of Plasticity., submitted to ICLR 2024.

- Attempts to explain the fundamental cause of the loss of plasticity in terms of trainability.
- Found that the curvature of loss surface explains the loss of plasticity (limited to synthetic experiments).

Bonus: Regenerative Regularization



Personal Research Direction

Research Direction I

- Empirically understand the loss of plasticity in terms of generalization.
  1. Existence of the loss of plasticity in a realistic scenario.
     - ResNet-18 with a chunked ImageNet.
  2. When does the loss of plasticity amplified?
     - Smaller data size for each chunk?
     - Noisy-label?
     - A larger distribution shift?
  3. Does existing methods alleviate / amplify the loss of plasticity?
     - Various architectural designs (e.g.,
     - Various regularization methods (e.g., dropout, L2, data augmentation).
     - Various plasticity-preserving methods (e.g., regenerative reg, feat.rank reg, shrink & perturb, reset).
  4. Propose method to mitigate the loss of plasticity (TBD).
     - Layer-wise regenerative regularization? Layer-wise gradient noise?

Recommended Readings

General

- On Warm-Starting Neural Network Training., NeurIPS 2020.
- Understanding plasticity in neural networks., ICML 2023.
- Loss of Plasticity in Deep Continual Learning., CoLLA 2022 talk.
- Continual Learning as Computationally Constrained Reinforcement Learning., COLLA 2023 talk.
- Maintaining Plasticity in Continual Learning via Regenerative Regularization., TMLR 2023.
- A study on the plasticity of neural networks., arXiv 2023.10.
- Curvature explains Loss of Plasticity., submitted to ICLR 2024.

Reinforcement Learning

- Understanding and Preventing Capacity Loss in Reinforcement Learning., ICLR 2022.
- The Primacy Bias in Deep Reinforcement Learning., ICML 2022.
- Sample-Efficient Reinforcement Learning by Breaking the Replay Ratio Barrier., ICLR 2023.
- Loss of Plasticity in Continual Deep Reinforcement Learning., TMLR 2023.
- Deep Reinforcement Learning with Plasticity Injection., NeurIPS 2023.
- Improving Input and Label Plasticity for Sample Efficient Reinforcement Learning., NeurIPS 2023. ← My paper!